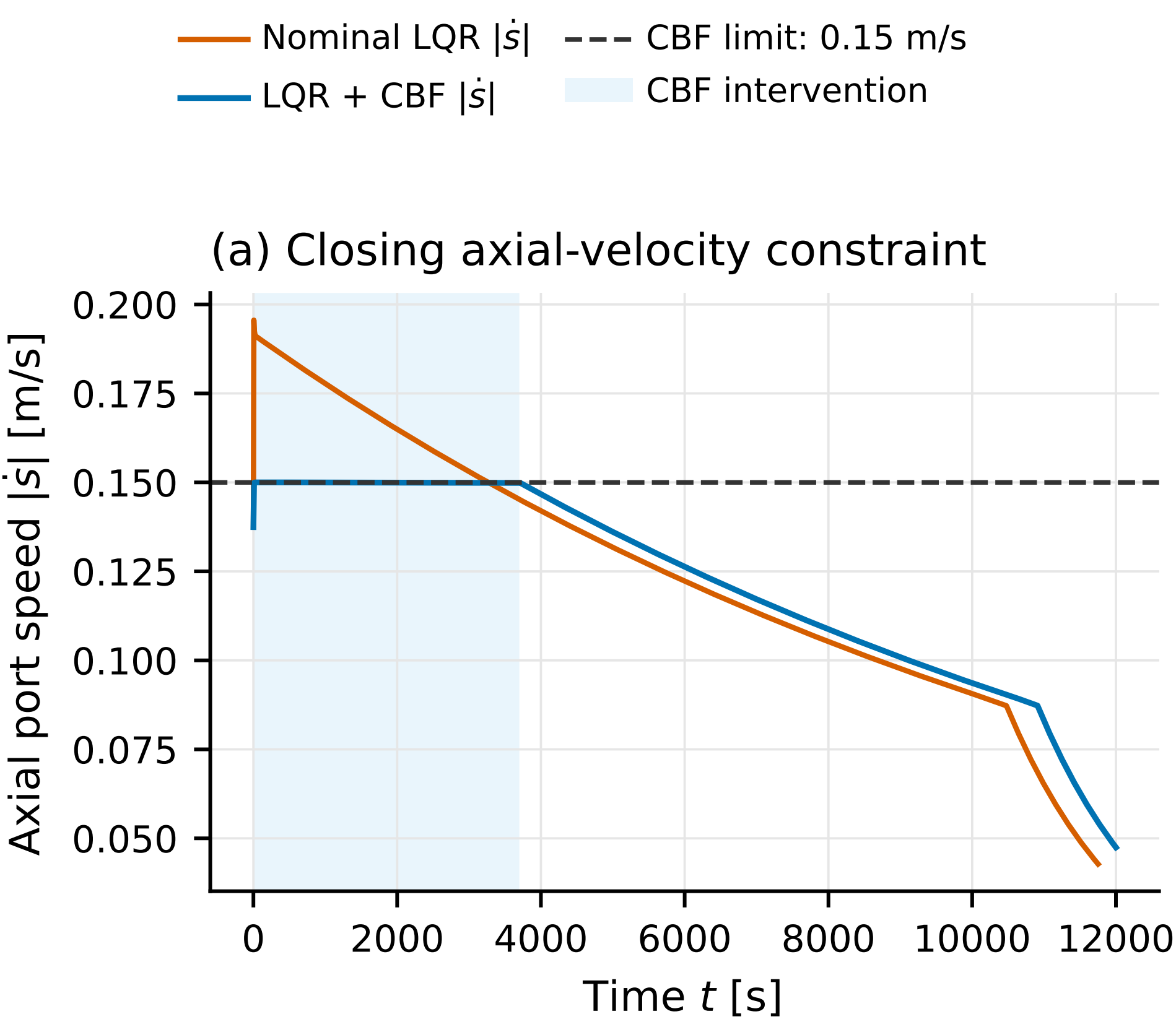

Nominal LQR |ṡ|
LQR + CBF |ṡ|
CBF limit: 0.15 m/s
CBF intervention
(a) Closing axial-velocity constraint
Axial port speed |ṡ| [m/s]
0.200
0.175
0.150
0.125
0.100
0.075
0.050
0
2000
4000
6000
8000
10000
12000
Time t [s]